MILITARY SPECIFICATION

MINOL-2 COMPOSITION

1. SCOPE

1.1 Scope.-This Military Specification covers Minol-2 Composition for use in Bombs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-248 - Trinitrotoluene (TNT)
MIL-A-512 - Aluminum Powder, Flake, Grained, and Atomized

PURCHASE DESCRIPTIONS

PA-PD-3087 - Ammonium Nitrate Grained (For Use in Explosives)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Inspection Terms and Definitions

FSC: 1376

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PUBLICATIONS

American Society for Testing and Materials

ASTM Designation E203-62 - Water Using Karl Fischer Reagent, Test for

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials

3.1.1 Ammonium Nitrate.-The ammonium nitrate shall comply with the requirements of Purchase Description PA-PD-3087. The moisture of the ammonium nitrate shall not exceed 0.15 percent when tested as specified in 4.2.2.

3.1.2 Aluminum Powder.-The aluminum powder shall comply with the requirements of Specification MIL-A-512, Type III, Grade F, Class 7.

3.1.3 Trinitrotoluene (TNT).-The TNT shall comply with the requirements of Specification MIL-T-248, Type I.

3.2 Composition.-The Minol-2 composition shall consist of a homogeneous mixture of the ingredients specified in Table I.

TABLE I

Ingredient	Percent
Ammonium Nitrate	40 ± 2
Aluminum Powder	20 ± 2
TNT	40 ± 2

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3.3 Moisture.-The moisture content of the Minol-2 shall not exceed 0.15 percent when tested as specified in 4.3.2.

3.4 Specific gravity.-The specific gravity of the Minol-2 shall not be less than 1.65 when tested as specified in 4.3.3.

3.5 Minol-2 particles (chips) used as filler.-Broken chips of Minol-2 shall not be less than 3/8 inch. The smaller size particles shall be removed by sieving through a 3/8 inch mesh sieve.

3.6 Workmanship.-The Minol-2 composition shall be free of all foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

NOTE: The ammonium nitrate shall be tested within 24 hours prior to loading.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification shall be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period.

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4.2.2 Ammonium nitrate sample for moisture determination.-Prior to loading the ammonium nitrate into the kettle, select five 25-gram samples from five different bags of ammonium nitrate. These samples shall be used for moisture determination in accordance with ASTM Method E203-62 except that a 7 to 9 gram sample shall be added to a 1:1 mixture of pyridine and methanol. If any of the samples fails to comply with the requirements of 3.1.1, all the ammonium nitrate which the sample represents shall be dried.[1]

4.2.3 Sampling for test shall be in accordance with Table II. If any of the samples fails to comply with the requirements of paragraph 3.2 through 3.6, the production which the samples represent shall be rejected. The samples shall be selected at random from the kettles and rods.

TABLE II

Test	Samples from kettle from each line	Samples of rods from each line	Samples of Chips from each line	Rejection of production samples
Composition	50 grams	50 grams		represents day's production
Moisture	50 grams	2 rods	50 grams	8 hours
Specific gravity	(see 4.3.3.1)	3 rods[2]		reject the 100 bombs
Workmanship	above samples	above samples	above samples	8 hours' production.

1 The ammonium nitrate shall be tested within twenty-four hours prior to loading.

2 The specific gravity of the rods shall be 1.65 on advisory basis.

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4.3 Procedures

4.3.1 Composition analysis, Major Defect, Code No. 01001, (See 6.3).-Add a 1 gram sample to a 250 ml beaker with 100 ml of acetone. Dissolve the sample in the acetone by stirring with a stirring rod. After the sample has dissolved, pour the residue in the beaker into a tared 50 ml. fritted glass crucible and aspirate with air. Dry the crucible in an oven at 100 ± 5 degrees C for one hour. Remove the crucible from the oven and place in a desiccator to cool. After the crucible is cool, weigh the crucible and contents and record the weight. Place the crucible and its contents on an aspirating assembly. While aspirating, pour hot water into the crucible until all the ammonium nitrate is dissolved. Then place the crucible and its contents into an oven at 100 ± 5 degrees C for one hour. Remove the crucible from the oven and place in a desiccator to cool. Remove from the desiccator and weigh the crucible with its contents and record the weight. Calculate the loss in weight as the percent ammonium nitrate. Calculate the gain in crucible weight as the percent of aluminum. Add the weight of the aluminum and ammonium nitrate and subtract from total weight of the sample. Calculate the remaining weight as the percent TNT.

4.3.2 Moisture, Major Defect, Code No. 02001.-The moisture shall be determined in accordance with ASTM Method E203-62 except that 8 to 10 grams samples shall be added to methanol.

4.3.3 Specific gravity.

4.3.3.1 Specific gravity of Minol-2 in bombs, Major Defect, Code No. 03001.-For determination of specific gravity of the charge, the inspector shall select and identify two bombs. The overall specific gravity of the charge in each sample bomb shall be determined as follows:

Weigh each sample bomb empty. Fill each sample bomb with water and weigh, making allowance for the weight of the water displaced by the components (e.g., auxiliary booster sleeve, auxiliary booster, adapter-booster) which are added during the loading. Empty each bomb and thoroughly dry. Place these sample bombs in the loading line at the rate of approximately two bombs per hundred and fill under conditions identical in all respects with those employed in filling regular bombs. Weigh each sample bomb, making allowance for the weight of the components which were added during loading. Calculate the specific gravity as follows:

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$$\text{Specific gravity} = \frac{Wt - We}{Ww - We}$$

where:

We = weight of an empty bomb
Ww = corrected weight of water-filled bomb
Wt = corrected weight of loaded bomb

4.3.3.2 Specific gravity of Minol-2 in rods.-The rods shall be weighed in air and the weight shall be recorded. Then the rod shall be weighed in water and the weight shall be recorded. The volume of the rod shall be determined by subtracting the weight of the rod in water from the weight of the rod in air. The specific gravity shall be determined as follows:

$$\text{Specific gravity} = \frac{\text{weight of Minol-2 rod}}{\text{volume of Minol-2 rod}}$$

4.3.4 Workmanship, Minor defect, Code No. 0'001.-A visual examination shall be made of the Minol rods.

5. PREPARATION FOR DELIVERY (Not Applicable).

6. NOTES

6.1 Intended use.-This Minol-2 is intended for use in bombs.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Test results.-A copy of the test results should be forwarded to: Commanding Officer, ATTN: SMUPA-ND2, Picatinny Arsenal, Dover, New Jersey, 07801.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A012

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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